

Product Requirements and Specification Document

Project Name

DataSnap - Visual Data Insights

Description

DataSnap is a web-based educational tool enabling users to upload CSV files and instantly view simple visualizations (bar and pie charts). The project introduces students to backend data processing and frontend chart rendering, focusing on data analytics and web development fundamentals. It is open-source and built with Spring Boot and modern web technologies.

1. Goals & Objectives

Goal	Description
Educational Tool	Teach basics of backend data processing and frontend visualization
Simple Visualizations	Provide instant bar and pie charts from uploaded CSV data
Open-Source	Ensure code is accessible and modifiable for learning
Easy to Use	Minimal setup, intuitive interface

2. Target Users

User Type	Description
Students	Learning web development and data analytics
Educators	Demonstrating data processing fundamentals
Hobbyists	Exploring open-source data visualization

3. Functional Requirements

3.1 User Stories

ID	As a...	I want to...	So that...
US1	User	Upload a CSV file	I can visualize my data
US2	User	View bar and pie charts	I can understand my data easily
US3	User	Download sample CSV	I can try the tool without my own data
US4	User	See error messages for bad uploads	I know when something goes wrong

3.2 Features

Feature	Description
CSV Upload	Upload CSV files via web interface
Data Parsing	Backend parses and validates CSV
Chart Rendering	Display bar and pie charts for selected columns
Sample Data	Provide downloadable sample CSV
Error Handling	Informative error messages for invalid files

4. Non-Functional Requirements

Requirement	Specification
Performance	Visualizations generated within 2 seconds for <1MB CSV
Usability	Simple, clean UI; minimal steps to visualize data
Security	No persistent storage; files processed in-memory only
Compatibility	Works on latest Chrome, Firefox, Edge
Accessibility	Basic keyboard navigation and screen reader support

5. Technical Specifications

5.1 Architecture Overview

```
[User] <-> [Frontend (React/HTML/JS)] <-> [Spring Boot Backend] <-> [CSV Processing]
```

5.2 Backend

Component	Technology	Responsibilities
API Server	Spring Boot	Handle file uploads, parse CSV, serve data
CSV Parser	OpenCSV/Commons	Parse and validate CSV files
Data API	REST	Provide parsed data for visualization

Endpoints:

- `POST /api/upload` — Accepts CSV file, returns parsed data or error
- `GET /api/sample` — Returns sample CSV file

5.3 Frontend

Component	Technology	Responsibilities
UI Framework	React/HTML/CSS	User interface, file upload, chart display
Chart Library	Chart.js/Recharts	Render bar and pie charts

Key UI Elements:

- File upload form
- Chart selection (bar/pie)
- Chart display area
- Error/message display

6. Data Handling

Step	Description
Upload	User uploads CSV via frontend
Validation	Backend checks file type, size, and structure
Parsing	Backend parses CSV, extracts columns and data
Response	Backend returns JSON with data/structure for charts
Visualization	Frontend renders charts based on user selection

7. Security & Privacy

- No data persistence; all files processed in-memory
- Limit file size (max 1MB)
- Accept only `.csv` files
- Sanitize all inputs

8. Open Source & Documentation

Item	Specification
License	MIT or Apache 2.0
Documentation	README with setup, usage, and contribution guidelines
Sample Data	Included in repository

9. Success Metrics

Metric	Target
Time to Visualization	<2 seconds for <1MB CSV
Usability Score	>80% positive feedback in user testing
Code Accessibility	100% code and docs public on GitHub

10. Out of Scope

- Advanced chart types (scatter, line, etc.)
- User authentication or accounts
- Persistent data storage
- Mobile app version

11. Milestones

Milestone	Description	Target Date
MVP	CSV upload, basic charts	Week 2
Error Handling	Robust validation, error messages	Week 3
Documentation	Complete README, sample data	Week 4

12. Appendix

Sample CSV Format:

```
Category,Value
A,10
B,20
C,30
```

End of Document